

Lecture 20

4th Nov '25

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} p & s \\ r & t \end{bmatrix}$$

$$\begin{bmatrix} a & b & 0 & 0 \\ c & d & 0 & 0 \\ 0 & 0 & a & b \\ 0 & 0 & c & d \end{bmatrix} \begin{bmatrix} e \\ f \\ g \\ h \end{bmatrix} = \begin{bmatrix} p \\ r \\ s \\ t \end{bmatrix}$$

$$L: \begin{bmatrix} a & a \\ a & a \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} a(e+f) \\ a(e+f) \end{bmatrix} \quad \textcircled{1} \text{ multiplication}$$

$$B: \begin{bmatrix} a & a \\ -a & -a \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} a(e+f) \\ -a(e+f) \end{bmatrix} \quad \textcircled{1} \text{ multiplication}$$

$$Y: \begin{bmatrix} a & 0 \\ a-b & b \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} ae \\ ae + b(f-c) \end{bmatrix} \quad \textcircled{2} \text{ multiplication}$$

$$J: \begin{bmatrix} a & b-a \\ 0 & b \end{bmatrix} \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} a(c-f) + bf \\ bf \end{bmatrix} \quad \textcircled{2} \text{ multiplication}$$

$$B = \begin{bmatrix} b & b & 0 & 0 \\ b & b & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & c & c \\ 0 & 0 & c & c \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 & 0 \\ c-b & 0 & 0 & c-b \\ b-c & 0 & 0 & b-c \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad E = \begin{bmatrix} a-b & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ c-b & 0 & a-c & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & a-b & 0 & b-c \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & a-c \end{bmatrix}$$

$$T(n) = 7T(n/2) + O(n^2)$$

Strassen's Algorithm

$$P = a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$$

$$P_1 = p_0 + p_1 x + \dots + p_{n/2-1} x^{n/2-1}$$

$$P_2 = p_{n/2} + p_{n/2+1} x + \dots + p_{n-1} x^{n/2-1}$$

$$P = P_1 + x^{n/2} P_2$$

$$Q = Q_1 + x^{n/2} Q_2$$

$$PQ = P_1 Q_1 + (P_1 Q_2 + P_2 Q_1) x^{n/2} + P_2 Q_2 x^n$$

1	2	3	...	n-1	n
$p(1)$	$p(2)$	$p(3)$	...	$p(n-1)$	$p(n)$
$q(1)$	$q(2)$	$q(3)$	...	$q(n-1)$	$q(n)$

Evaluate degree $n-1$ polynomial at n points
 (for $2n-1$ evaluation just pad zeroes)

Eg:

$$P(x) = 2x^2 + x - 1$$

$$Q(x) = 3x^2 - x + 1$$

Evaluatⁿ has to be done at 5 points
 degⁿ of product of polynomials is 4

$$P(x) = 0x^4 + 0x^3 + 2x^2 + x - 1$$

evaluate at 5 points

Choosing points of evaluatⁿ of efficient calculation

→ Reuse of computation

$$P(x) = \underbrace{\sum_{i=0}^{n/2-1} \underbrace{a_{2i}}_{P_e} x^{2i}}_E + \underbrace{\sum_{i=0}^{n/2-1} \underbrace{a_{2i+1}}_{P_o} x^{2i+1}}_O$$

$$P(x) = E + O$$

$$P(-x) = E - O$$

↓ polynomial of degree $n/2$
 (substitute $x^2 = t$)

$$P(x) = P_e(x^2) + x \cdot P_o(x^2)$$

We take following pts

$$x_1$$

$$x_2$$

$$\vdots$$

$$\vdots$$

$$x_{n/2}$$

$$-x_1$$

$$-x_2$$

$$\vdots$$

$$\vdots$$

$$-x_{n/2}$$

$$\left[\begin{array}{c} P_e \quad \& \quad P_o \\ \hline x_1^2 \quad x_2^2 \quad \dots \quad x_{n/2}^2 \end{array} \right] \text{ put of evaluator for } P_e \& P_o$$